

# JANAK SHARDA

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## SUMMARY

- Advanced package design and developing **CAD automation tools for package design**, account for different electrical and packaging requirements.
- Developing **evaluation frameworks for large-scale systems that perform LLM inference / training accelerators**, focusing on power, performance, and thermal studies to identify various bottlenecks.
- Software-hardware co-design of **hardware accelerator for large language models (LLMs)** using emerging 3D DRAM memory, advanced packaging techniques like 2.5D/3D integration and co-packaged optics.
- Implementation of **thermal-aware near-sensor computation-based 3D stacked CMOS Image sensor hardware for autonomous driving**.
- **Best B.Tech project award** in Electrical Engineering Department, IIT Delhi, for designing compute-in-memory based hardware accelerator.
- Ph.D. in hardware accelerators, heterogenous integration, co-packaged optics, compute-in-memory, software-hardware co-design, thermal analysis etc.

## EDUCATION

| Year             | Degree/Exam                           | Institute                                       | CGPA            |
|------------------|---------------------------------------|-------------------------------------------------|-----------------|
| <b>2021-2026</b> | <b>Ph.D., Electrical Engineering</b>  | <b>Georgia Institute of Technology, Atlanta</b> | <b>3.42/4</b>   |
| 2017-2021        | <b>B.Tech, Electrical Engineering</b> | <b>Indian Institute of Technology, Delhi</b>    | <b>9.034/10</b> |

## PUBLICATIONS

- J. Sharda *et. al.*, **Training Trillion-Parameter LLMs with 3D DRAM-based Accelerator and Co-Packaged Optical Interconnects**, *IEEE T-CPMT*
- J. Sharda *et. al.*, **System-Technology Co-Optimization Framework for LLM Accelerators with Advanced Packaging**, *IEEE JETCAS*, 2025.
- Y.-C. Luo\*, A. Lu\*, J. Sharda\* *et. al.*, **Thermally Constrained Codesign of Heterogeneous 3-D Integration of Compute-in-Memory, Digital ML Accelerator, and RISC-V Cores for Mixed ML and Non-ML Workloads**, *IEEE T-VLSI*, 2024.
- J. Sharda *et. al.*, **Design and Thermal Analysis of 2.5D and 3D Integrated System of a CMOS Image Sensor and a Sparsity-Aware Accelerator for Autonomous Driving**, *IEEE J-EDS*.
- J. Sharda\*, W. Li\* *et. al.*, **Temporal Frame Filtering for Autonomous Driving Using 3D-Stacked Global Shutter CIS With IWO Buffer Memory and Near-Pixel Compute**, *IEEE TCAS-I*, 2023.
- J. Sharda *et. al.*, **A Crossbar Array of Analog-Digital-Hybrid Volatile Memory Synapse Cells for Energy-Efficient On-Chip Learning**, *IEEE ISCAS*, 2021.
- D. Kaushik *et. al.*, **Synapse cell optimization and back-propagation algorithm implementation in a domain wall synapse based crossbar Neural Network for scalable on-chip learning**, *IOP Science Nanotechnology*, 2020.

Full list of publications: Janak Sharda - Google Scholar

\*these authors equally contributed in the paper

## TECHNICAL SKILLS

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**Computer Languages** C/C++, Python, MATLAB, JAVA, Assembly Language(SimpleRisc), Verilog HDL  
**Software & Tools** Cadence Virtuoso, Cadence Allegro, Ansys Mechanical APDL, Ansys HFSS, SUPREM

## EXPERIENCE

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**Package Design Engineer (Feb'26-Present):** Apple, Cupertino, CA

- Design advanced packages and CAD flow improvement

**Package Design Intern (May'25-Aug'25):** Apple, Cupertino, CA

- Developed CAD automation tool for component auto placement in package with > 300 components
- Account for various electrical and packaging needs specific to the module while performing placement.
- Potentially reduce package design effort and obtain improvement in overall design metrics.

**Advanced Package Research Engineer Intern (May'24-Aug'24):** Intel Corporation, Chandler, AZ

- Worked on designing hardware accelerators for LLMs using advanced packaging techniques such as 2.5D/3D integration
- Performed system-technology co-optimization for systems with different packaging schemes for the Llama2-13B model
- Integrated with an optimization algorithm to evaluate the optimal dataflow for a given LLM architecture and hardware constraints for different packages

**Graduate Research Assistant(Jan'21-Now):** Prof. Shimeng Yu, Dept. of ECE, Georgia Tech

- **Accelerator design for Large Language Models**
- Software-hardware co-design of hardware accelerator for large language models using **3D stackable DRAM and heterogenous integration** of logic and memory.
- Proposed a **layer-wise sparsity quantization hybrid** technique and performed software-hardware co-design to reduce the energy consumption.
- Achieved a **throughput of 160k tokens/s, energy efficiency of 25 TOPS/W and an area efficiency of 14 TOPS/mm<sup>2</sup>**
- Extended the work to a large-scale cluster of such accelerators capable of performing training of 1-trillion parameter model using co-packaged optics
- **CMOS Image Sensor for Autonomous Driving**
- Designed **Thermal-aware 3D stacked CMOS Image Sensor** for multi-object tracking for **autonomous driving**.
- Explored 2.5D/3D integration of CIS and accelerator for accurate inference of Resnet-50 based QDTrack network.
- Designed **heterogenous 3D integration** of photodiodes with **IWO-FETs** based buffer memory and near-pixel compute circuits using **Cu-Cu hybrid bonding** and obtained **global shutter** operation of CIS.
- Simulated different components and obtained **45.8 TOPS/W** energy efficiency and **0.12 TOPS/mm<sup>2</sup>** area efficiency.
- Modelled complete system in **ANSYS mechanical** for thermal simulations to obtain the peak temperature.
- **Thermal-aware framework for Mixed workloads for AR/VR**
- Designed a **thermal-aware framework** for operating points for a heterogenous accelerator comprising of **compute-in-memory, RISC-V cores, and digital accelerator**.
- The framework optimizes for power and latency to determine the operating point of different compute blocks and air cooling and keeps the peak temperature within specified limit.

**Graduate Teaching Assistant(Sep'21-Dec'21):** Prof. Nivedita Bhattacharya, Dept. of ECE, Georgia Tech

- Teaching Assistant for a course on VLSI Design(ECE3150), responsible for grading and handling project doubts.

## ACADEMIC ACHIEVEMENTS

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- Awarded **Best Bachelor's thesis award** in Electrical Engineering Department, IIT Delhi.
- Awarded **Summer Undergraduate Research Award** for the work done under Prof. Debanjan Bhowmik.

- Won **IIT Delhi Semester Merit Award** in both 1st and 2nd semester 2017-18 (given to **top 7%** of all the students).
- Secured an **All India Rank 410** in Joint Entrance Exam Advanced 2017 among 200 thousand candidates.
- Qualified **KVPY** exam, by Indian Institute of Science, Bangalore by securing an **All India Rank 126**.
- Ranked in **top 0.01%** among 2 million candidates appearing in JEE mains - 2017.
- Ranked in **top 1%** in India in **NSEC 2017** (National Standard Examination in Chemistry) and **NSEP 2017** (National Standard Examination in Physics).

## PROJECTS

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**In-Memory Computation based Hardware Accelerator using Transistors and Spintronics(Feb'19-May'21):**  
Prof. D. Bhowmik, Dept. of EE, IIT Delhi

- Implemented hardware accelerators for CNNs using **transistors** and **spintronics** synapse based analog crossbars.
- Designed separate crossbars for MSBs and LSBs, to efficiently implement and got **70% accuracy** on CIFAR10 dataset.
- Devised a thresholding scheme and synapse cell based on MOSFETs for Analog crossbar based deep neural networks.
- Improved speed, area, power, decay rate and done mismatch analysis of volatile synapses, trained it on MNIST.
- Designed transistor based circuit for Spintronic based Non-Volatile Synapse Cell and tested it on MNIST Dataset.

**12-bit Pipelined SAR ADC(Jan'22-Apr'22):**

Prof. S. Li, Dept. of ECE, Georgia Tech

- Designed a 12-bit pipelined SAR ADC using an 8-bit SAR ADC in the first stage and 5-bit SAR ADC in second stage.
- Designed various components like high-gain cascode OTA, SAR logic, DAC and obtained power consumption of **1mW**.

**Fabricating an IC in Cleanroom(May'22-Aug'22):**

Prof. A. Frazier, Dept. of ECE, Georgia Tech

- Fabricating and characterizing an IC consisting of individual MOSFETs, inverters, resistors and ring oscillators .
- Cleanroom experience of various processes such as photolithography, diffusion, oxidation, etching, metallization etc..

**Characterization of Railway Track Vibration(Sept'18-Jan'19):**

Prof. S.D.Joshi, Dept. of EE, IIT Delhi

- Designed and employed a sensor capable of collecting **real time data** of track vibrations for normal running of train.
- Characterized different types of vibrations for different track conditions using standard signal processing like autocorrelation, power spectral density and machine learning techniques like Neural Network, SVM etc.

**Detecting damages in vertebral column(Jan'20-Mar'20):**

Prof. A.P. Prathosh, Dept. of EE, IIT Delhi

- UNet based Image segmentation of X-Ray images of different regions of Vertebral Column and obtained 0.7 Dice Score.
- Trained a squeezeNet based classifier and classified X-Ray images as damaged or normal with validation score of 72%.

**SimpleRisc processor with 5 stage pipeline(Sep'19-Nov'19):** Prof. Smruti R.Sarangi, Dept. of CS, IIT Delhi

- Implemented a 5-stage pipeline processor for SimpleRisc Assembly language instructions.
- Implemented forwarding, data locks and branch locks to check for control and data hazards.

**Small Search Engine (Sept'18-Oct'18):**

Prof. Amitabha Bagchi, Dept. of CS, IIT Delhi

- Implemented Search Engine using **Inverted Indexing** for a set of webpages using **Hash Table**.
- Devised connector words skipping mechanism for faster search.

## RELEVANT COURSES

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**Electrical:** Microelectronic System Packaging(ECE6776), Analog Integrated System Design(ECE6414), Memory Devices and Technology(ECE8803), Advance Machine Learning(ELL888), Advanced VLSI systems(ECE6130), Hardware for Machine Learning (ECE8803), Generative and Geometric Deep Learning(ECE 8803)

**Computer Science:** Systems for Machine Learning(CS 8803), Data Structures and Algorithms(COL106), Systems in Machine Learning(CS8803)

**Mathematics:** Math. Foundations for Machine Learning(ECE7750), Probability and Stochastic Processes(MTL106)

**Laboratories:** IC Fabrication Lab(ECE4452)